PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 1 Submission

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Topic: Unit 1 - Lab 1

Digital Forensics and Incident Responses

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Task 1: Data Acquisition using dd

In digital forensics, data acquisition is very important in the process of investigation of digital evidence. One of the most widely used tools for acquiring raw data from storage devices, partitions dd and RAM is the command. This tool is very effective in creating exact forensic images of storage devices

Task 1.1: Creating a Forensic Image

A forensic image is a bit-by-bit copy of the source media including everything like files, file systems and unallocated space. This image is in a or a file.

To make a forensic image of Flash Drive:
WARNING: Make sure that you double check the device that you are reading from and writing to. There is no GUI for this, so it is possible that you overwrite your existing Operating system. If you are not sure, then do not run the commands

Step1: Identify the device we will be making a forensic image of

1) Command: diskutil (for macos)

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil
Disk Utility Tool
Utility to manage local disks and volumes
Most commands require an administrator or root user

WARNING: Most destructive operations are not prompted

Usage:  diskutil [quiet] <verb> <options>, where <verb> is as follows:

     list                (List the partitions of a disk)
     info[rmation]       (Get information on a specific disk or partition)
     listFilesystems     (List file systems available for formatting)
     listClients         (List all current disk management clients)
     activity            (Continuous log of system-wide disk arbitration)

     u[n]mount           (Unmount a single volume)
     unmountDisk         (Unmount an entire disk (all volumes))
     eject               (Eject a disk)
     mount               (Mount a single volume)
     mountDisk           (Mount an entire disk (all mountable volumes))

     enableJournal       (Enable HFS+ journaling on a mounted HFS+ volume)
     disableJournal      (Disable HFS+ journaling on a mounted HFS+ volume)
     moveJournal         (Move the HFS+ journal onto another volume)
     enableOwnership     (Exact on-disk User/Group IDs on a mounted volume)
     disableOwnership    (Ignore on-disk User/Group IDs on a mounted volume)

     rename[Volume]      (Rename a volume)

     verifyVolume        (Verify the file system data structures of a volume)
     repairVolume        (Repair the file system data structures of a volume)
     verifyDisk          (Verify the components of a partition map of a disk)
     repairDisk          (Repair the components of a partition map of a disk)

     eraseDisk           (Erase an existing disk, removing all volumes)
     eraseVolume         (Erase an existing volume)
     reformat            (Erase an existing volume with same name and type)
     eraseOptical        (Erase optical media (CD/RW, DVD/RW, etc.))
     zeroDisk            (Erase a disk, writing zeros to the media)
     randomDisk          (Erase a disk, writing random data to the media)
     secureErase         (Securely erase a disk or freespace on a volume)

     partitionDisk       ((re)Partition a disk, removing all volumes)
     addPartition        (Create a new partition to occupy free space)
     splitPartition      (Split an existing partition into two or more)
     mergePartitions     (Combine two or more existing partitions into one)
     resizeVolume        (Resize a volume, increasing or decreasing its size)

     appleRAID <verb>    (Perform additional verbs related to AppleRAID)
     coreStorage <verb>  (Perform additional verbs related to CoreStorage)
     apfs <verb>         (Perform additional verbs related to APFS)
     image <verb>        (Perform additional verbs related to DiskImage)

diskutil <verb> with no options will provide help on that verb
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil list
/dev/disk0 (internal, physical):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        *500.3 GB   disk0
   1:             Apple_APFS_ISC Container disk1         524.3 MB   disk0s1
   2:                 Apple_APFS Container disk3         494.4 GB   disk0s2
   3:        Apple_APFS_Recovery Container disk2         5.4 GB     disk0s3

/dev/disk3 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +494.4 GB   disk3
                                 Physical Store disk0s2
   1:                APFS Volume Macintosh HD            16.4 GB    disk3s1
   2:              APFS Snapshot com.apple.os.update-... 16.4 GB    disk3s1s1
   3:                APFS Volume Preboot                 16.1 GB    disk3s2
   4:                APFS Volume Recovery                2.5 GB     disk3s3
   5:                APFS Volume Data                    372.1 GB   disk3s5
   6:                APFS Volume VM                      7.5 GB     disk3s6

pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil info
Usage:  diskutil info[rmation] [-plist]
        MountPoint|DiskIdentifier|DeviceNode|UUID | -all
Display detailed information about a disk or partition.
Root or administrator access is not required.
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil listFilesystems
Formattable file systems

These file system personalities can be used for erasing and partitioning.
When specifying a personality as a parameter to a verb, case is not considered.
Certain common aliases (also case-insensitive) are listed below as well.

-------------------------------------------------------------------------------
PERSONALITY                     USER VISIBLE NAME
-------------------------------------------------------------------------------
Case-sensitive APFS             APFS (Case-sensitive)
  (or) APFSX
APFS                            APFS
  (or) APFSI
ExFAT                           ExFAT
Free Space                      Free Space
  (or) FREE
MS-DOS                          MS-DOS (FAT)
MS-DOS FAT12                    MS-DOS (FAT12)
MS-DOS FAT16                    MS-DOS (FAT16)
MS-DOS FAT32                    MS-DOS (FAT32)
  (or) FAT32
HFS+                            Mac OS Extended
Case-sensitive HFS+             Mac OS Extended (Case-sensitive)
  (or) HFSX
Case-sensitive Journaled HFS+   Mac OS Extended (Case-sensitive, Journaled)
  (or) JHFSX
Journaled HFS+                  Mac OS Extended (Journaled)
  (or) JHFS+
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

After inserting usb:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil list
/dev/disk0 (internal, physical):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        *500.3 GB   disk0
   1:             Apple_APFS_ISC Container disk1         524.3 MB   disk0s1
   2:                 Apple_APFS Container disk3         494.4 GB   disk0s2
   3:        Apple_APFS_Recovery Container disk2         5.4 GB     disk0s3

/dev/disk3 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +494.4 GB   disk3
                                 Physical Store disk0s2
   1:                APFS Volume Macintosh HD            16.4 GB    disk3s1
   2:              APFS Snapshot com.apple.os.update-... 16.4 GB    disk3s1s1
   3:                APFS Volume Preboot                 16.1 GB    disk3s2
   4:                APFS Volume Recovery                2.5 GB     disk3s3
   5:                APFS Volume Data                    375.7 GB   disk3s5
   6:                APFS Volume VM                      5.4 GB     disk3s6

/dev/disk4 (external, physical):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:     FDisk_partition_scheme                        *4.0 GB     disk4
   1:               Windows_NTFS SONY_4GR               4.0 GB     disk4s1

pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

Step2: Use to make the forensic image of the device and store it dd in a file. Run the following command:

2) Command: sudo dd if=/dev/rdisk4 of=/Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd bs=64k status=progress

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil unmountDisk /dev/disk4
Unmount of all volumes on disk4 was successful
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo dd if=/dev/rdisk4 of=~/sony_usb_image.dd bs=64k status=progress
Password:
dd: ~/sony_usb_image.dd: No such file or directory
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo dd if=/dev/rdisk4 of=/Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd bs=64k status=progress
  4031840256 bytes (4032 MB, 3845 MiB) transferred 199.994s, 20 MB/s
61776+0 records in
61776+0 records out
4048551936 bytes transferred in 200.792997 secs (20162814 bytes/sec)
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

Step3: Once the process is complete, it's important to verify the integrity of the forensic image. This can be done by comparing the has values of the source device and the forensic image. UNIX can calculate the hash value of any file from the command line itself

3) Command: sudo shasum -a 256 /dev/rdisk4

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo shasum -a 256 /dev/rdisk4
Password:
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /dev/rdisk4
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo shasum -a 256 /dev/rdisk4
Password:
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /dev/rdisk4
```

hash: 3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5 /dev/rdisk4

4) Command: sudo shasum -a 256 /dev/rdisk4

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %shasum -a 256 /Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %shasum -a 256 /Users/pranavhemanth/Code/
Academics/DFIR-S6/sony_usb_image.dd
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /Users/pranavh
emanth/Code/Academics/DFIR-S6/sony_usb_image.dd
```

hash: 3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5 /Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd

Both hashes:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo dd if=/dev/rdisk4 of=/Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd bs=64k status=progress
  4031840256 bytes (4032 MB, 3845 MiB) transferred 199.994s, 20 MB/s
61776+0 records in
61776+0 records out
4048551936 bytes transferred in 200.792997 secs (20162814 bytes/sec)
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo shasum -a 256 /dev/rdisk4
Password:
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /dev/rdisk4
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %shasum -a 256 /Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image.dd
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil unmountDisk /dev/disk4
Unmount of all volumes on disk4 was successful
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo dd if=/dev/rdisk4 of=~/sony_usb_ima
ge.dd bs=64k status=progress
Password:
dd: ~/sony_usb_image.dd: No such file or directory
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo dd if=/dev/rdisk4 of=/Users/pranavh
emanth/Code/Academics/DFIR-S6/sony_usb_image.dd bs=64k status=progress
  4031840256 bytes (4032 MB, 3845 MiB) transferred 199.994s, 20 MB/s
61776+0 records in
61776+0 records out
4048551936 bytes transferred in 200.792997 secs (20162814 bytes/sec)
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sudo shasum -a 256 /dev/rdisk4
Password:
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /dev/rdisk4
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %shasum -a 256 /Users/pranavhemanth/Code/
Academics/DFIR-S6/sony_usb_image.dd
3608595324d8b399ed3fc12b86fa7769a44ae2bac1d2c37fb7e4c90821c85ed5  /Users/pranavh
emanth/Code/Academics/DFIR-S6/sony_usb_image.dd
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

Sony_usb_image.dd

```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %ls -lh sony_usb_image.dd
-rw-r--r--  1 root  staff   3.8G Jan 27 09:10 sony_usb_image.dd
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %ls -l@ sony_usb_image.dd
-rw-r--r--  1 root  staff  4048551936 Jan 27 09:10 sony_usb_image.dd
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %
```

Task 1.2: Creating a Forensic Image of a partition

There are some scenarios where the entire drive need not be copied, and one partition is enough. can also be used in such cases, we just dd have to change the input file that we are using.
UNIX treats all device drivers and hardware as files and we can use this to our advantage. Files like is the entire disk itself, dev/sda but the file is a partition of the disk. In order to create dev/sda1 a forensic image of this partition, we can do this

Step1: Use to make the forensic image of the device partition and store it
dd in a file. Run the following command:

5) Command: sudo dd if=/dev/rdisk4s1 of=/Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image_partition.dd bs=64k status=progress

Explanation:
In certain forensic scenarios, acquiring an image of the entire storage device is unnecessary when the relevant evidence is contained within a specific partition. UNIX-based operating systems treat hardware devices and partitions as files, which allows forensic tools to interact directly with them.

In this task, instead of imaging the entire disk, only the required partition was acquired. While the whole disk is represented by a device file such as /dev/disk4, individual partitions are represented by separate device files such as /dev/disk4s1. By specifying the partition device file as the input, a bit-by-bit forensic image of only that partition can be created.

Before acquisition, the partition was unmounted to prevent any read or write operations during imaging. The dd utility was then used to create a forensic image of the partition by reading raw data from the partition device and writing it to an image file.

sudo dd if=/dev/rdisk4s1 of=/Users/pranavhemanth/Code/Academics/DFIR-S6/sony_usb_image_partition.dd bs=64k status=progress

This process resulted in a smaller and faster image acquisition while preserving all filesystem data contained within the selected partition.

Task 1.3: Capture Memory Dump

dd is not a tool that should be used to capture the memory dump, but it can be used as well. In this case we just have to target the right device file. In the case of the RAM the input file is dev/mem. So in order to make a Memory Dump, we can run the following command

Step1: Run memory dump in linux only (macos does not allow):

6) Command: sudo dd if=/dev/mem of=forensic_image.dd bs=64K status=progress

Explanation:
Memory acquisition is an important aspect of digital forensics, as volatile memory can contain valuable information such as running processes, encryption keys, and network connections. On UNIX-based systems, memory can be accessed using specific device files.

Although dd is primarily designed for disk acquisition and is not the preferred tool for memory forensics, it can be used on Linux systems by targeting the physical memory device file /dev/mem. By reading directly from this device file, a raw memory dump can be created.

However, modern macOS systems restrict access to physical memory for security reasons, and the /dev/mem interface is not available. Therefore, memory acquisition using dd is not supported on macOS and must be demonstrated on Linux systems only.

Linux only command:
sudo dd if=/dev/mem of=forensic_image.dd bs=64K status=progress

This command reads the contents of physical memory and stores it in a forensic image file for further analysis. On macOS, specialized tools with kernel-level access are required for memory acquisition instead of dd.

FTK Imager Walkthrough

Introduction to FTK Imager

FTK Imager is a GUI based tool that is used to make forensic images of Flash Drives, Disks, Partitions and Volatile Memory. You can preview the contents of the forensic images stored on the local machine or on the network drive. It can also mount an image and leverage the power of Windows File Explorer to see the content on the original drive as is.

Task 2.1: Installing FTK Imager

FTK Imager is downloaded from the website FTK Imager. Just go to the website and click on the Download button. This will take you to another website to fill in data. You don't have to give your real information. Use 33mail or 10minutemail to create a dummy account. It does not ask for verification. Once you launch the downloaded application, it will install FTK Imager into the path that you choose. Follow the steps displayed by the wizard and you will have FTK setup.

Task 2.2: Making a Forensic Image

With FTK Imager's "User Friendly" UI, you can easily make images of Physical Drives and Logical Drives. The difference between the two is quite simple. Physical Drive is the complete Physical Disk. It is the equivalent of /dev/nvme0n1 partition on Linux, while a Logical Drive is the the partition we created like C: or D:

Task 2.3: Physical Drive

Let's now connect a Flash Drive to the computer and make a Forensic Image of that USB

We see a list of Physical devices here and we choose the one that we want to make an image of. In this case SanDisk Cruzer Blade is the Flash Drive that we want to make an image of.

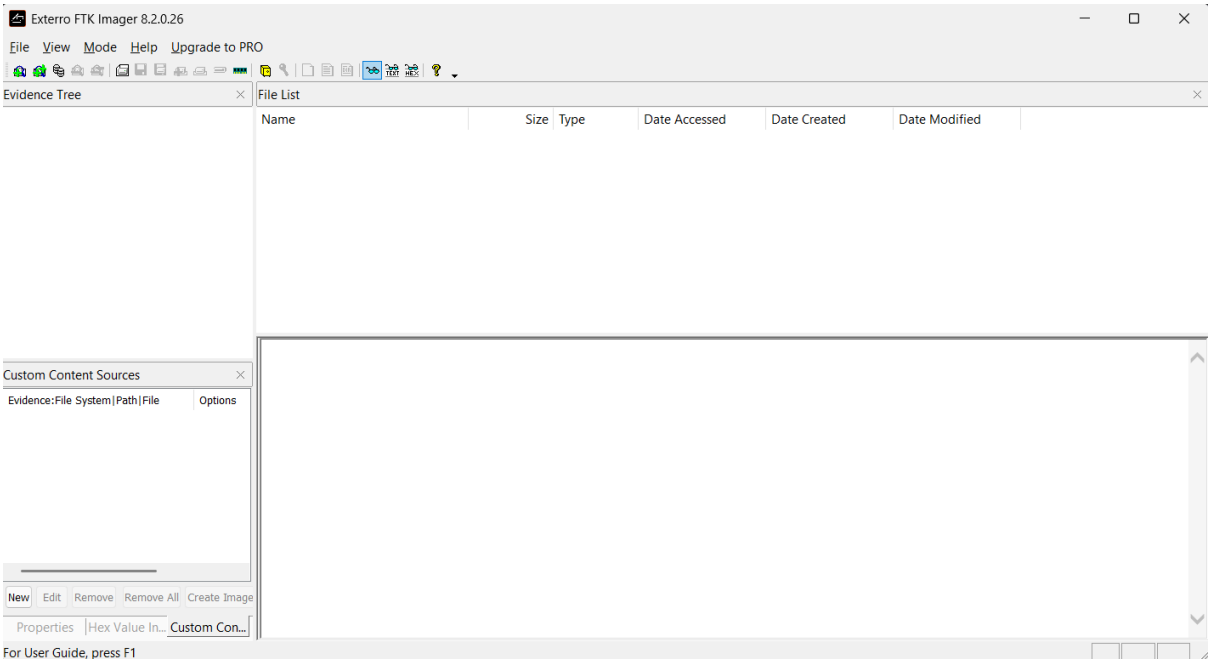
Select it, and then click on Finish.

This will lead us to another set of questions asking which format we want to use, where we want to save it, if we want to Verify after Imaging etc. Answer all these questions as per your need, and then click on Start.

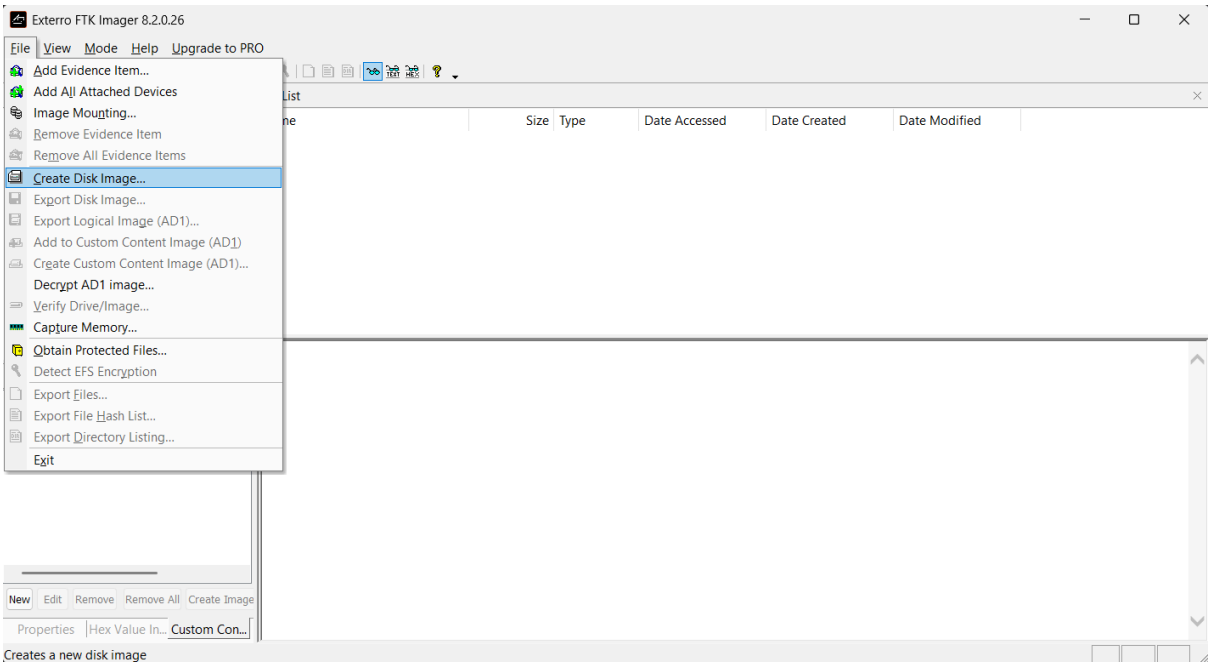
This is how it should be creating the Forensic Image. Once it is done, it will be stored in the folder of your choice

Once this is done, FTK Imager will display the results of the verification of the forensic image and compare them with the original. The images having the same Hash Values will be labeled as if the two In the same directory as the forensic images, there will be a text file which contains the details about the forensic image. It contains information about the media, file hashes and details that you entered while creating the image.

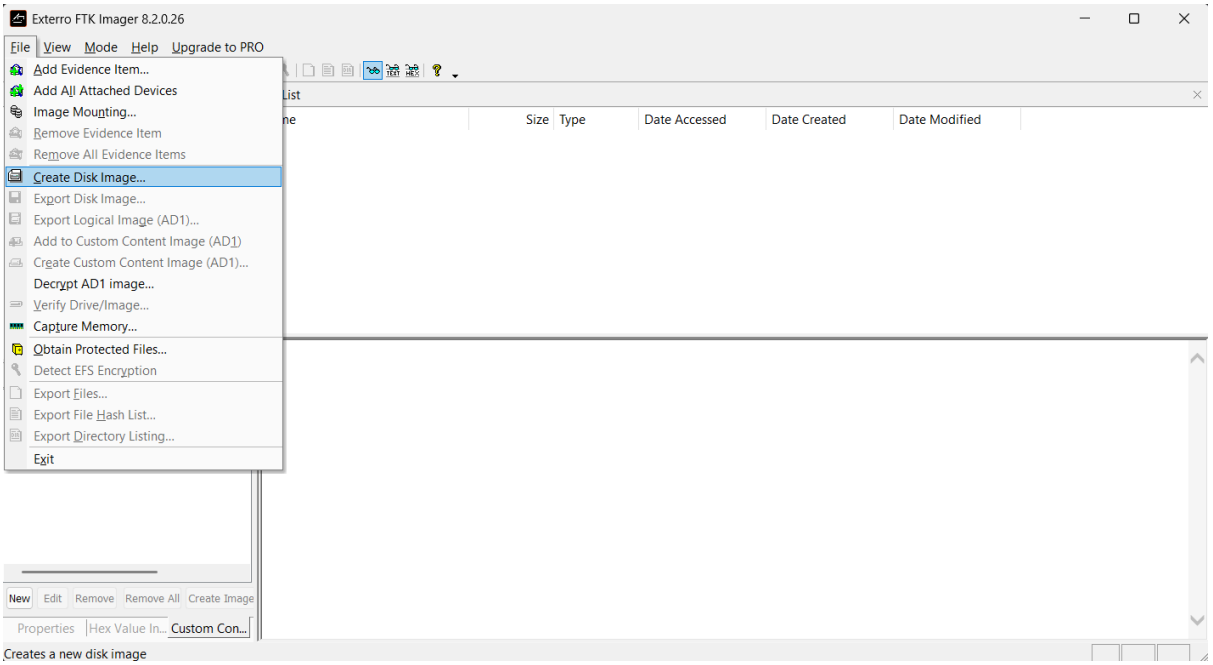
3.1 Launching FTK Imager and Creating Disk Image



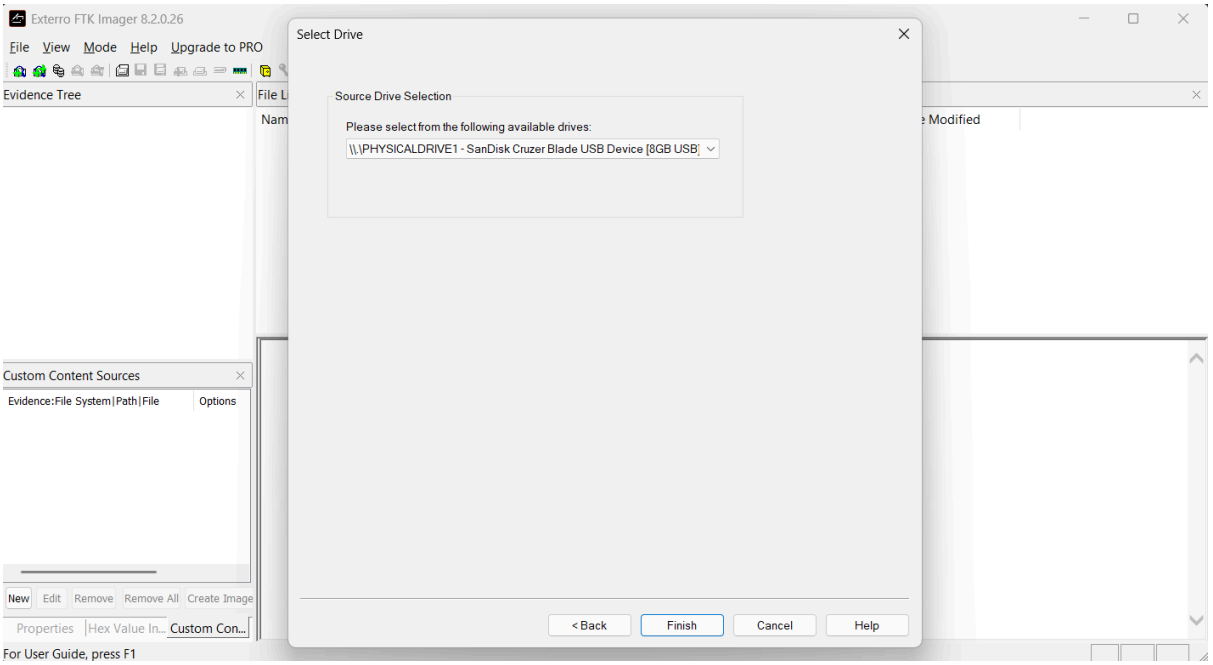
Now, to create a Disk Image, click on File > Create Disk Image.

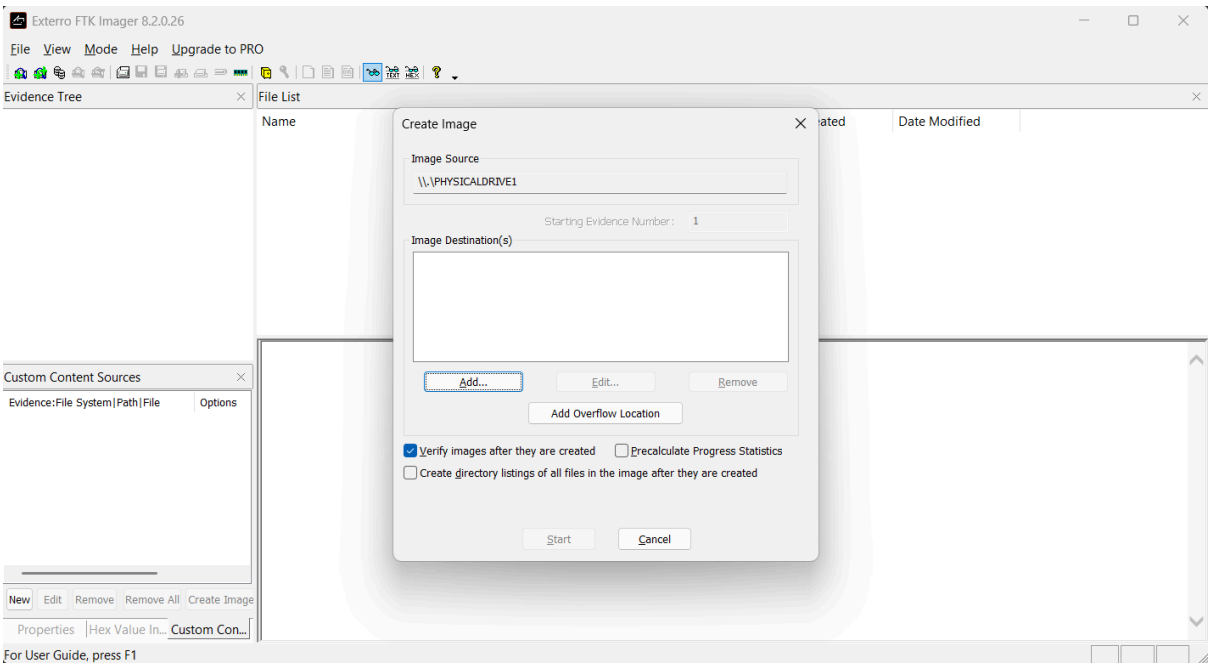


You can choose the source based on the drive you have. It can be a physical or a logical drive depending on your evidence.

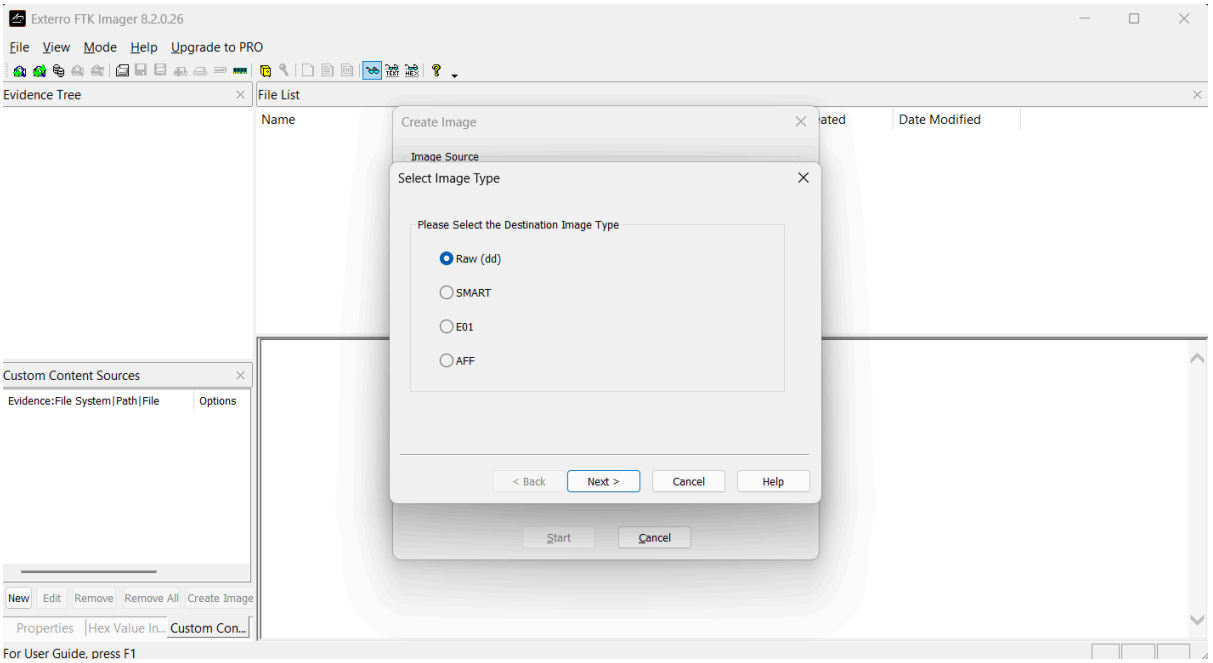


3.2 Selecting Physical Drive

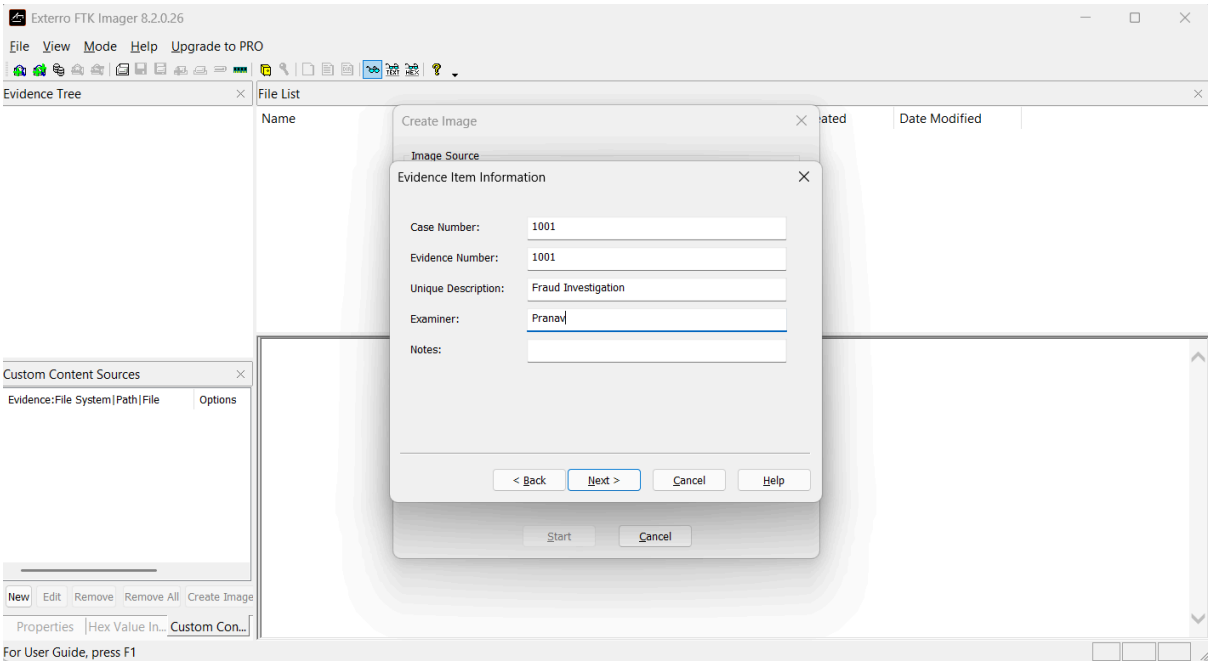




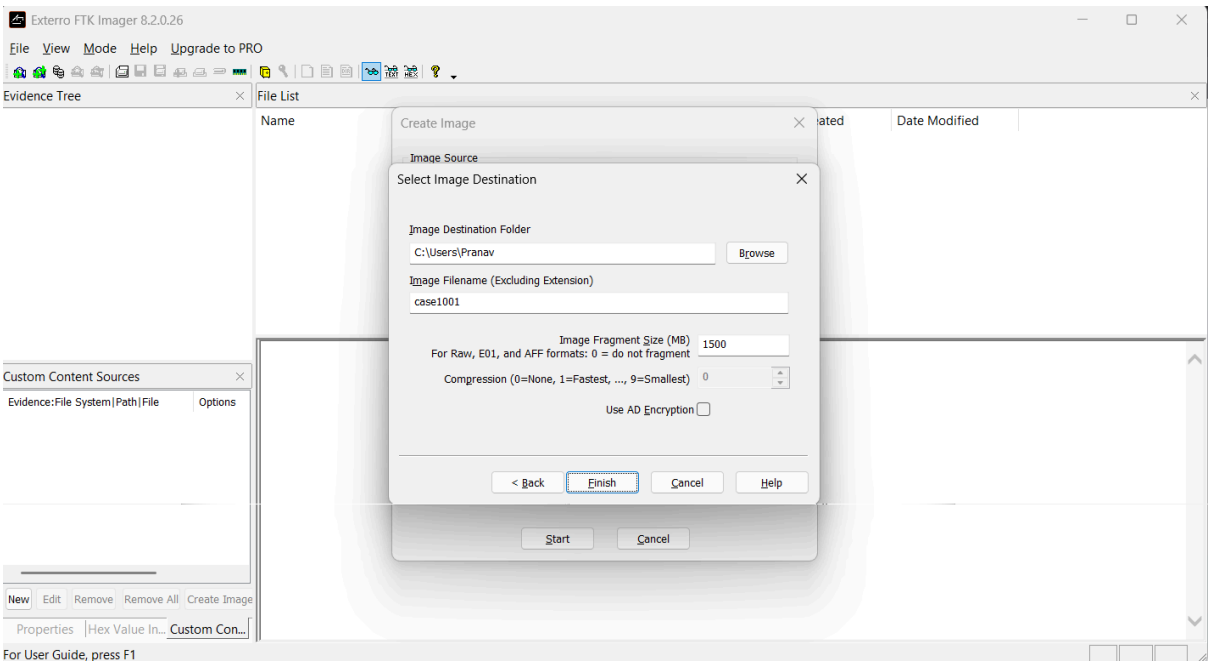
3.3 Selecting the Image Format



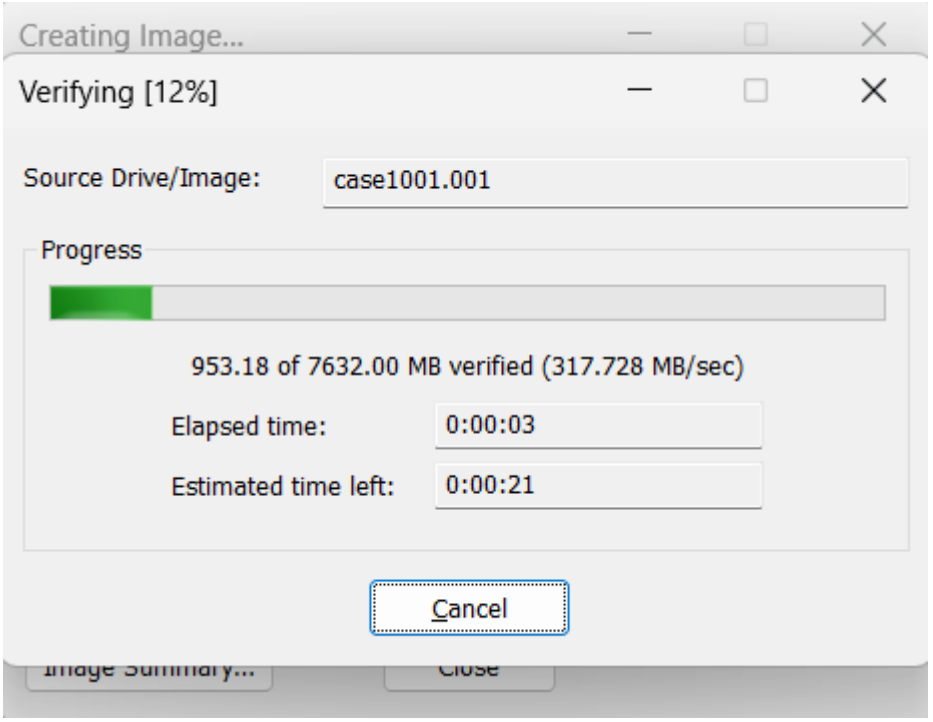
Add the details of the image to proceed.



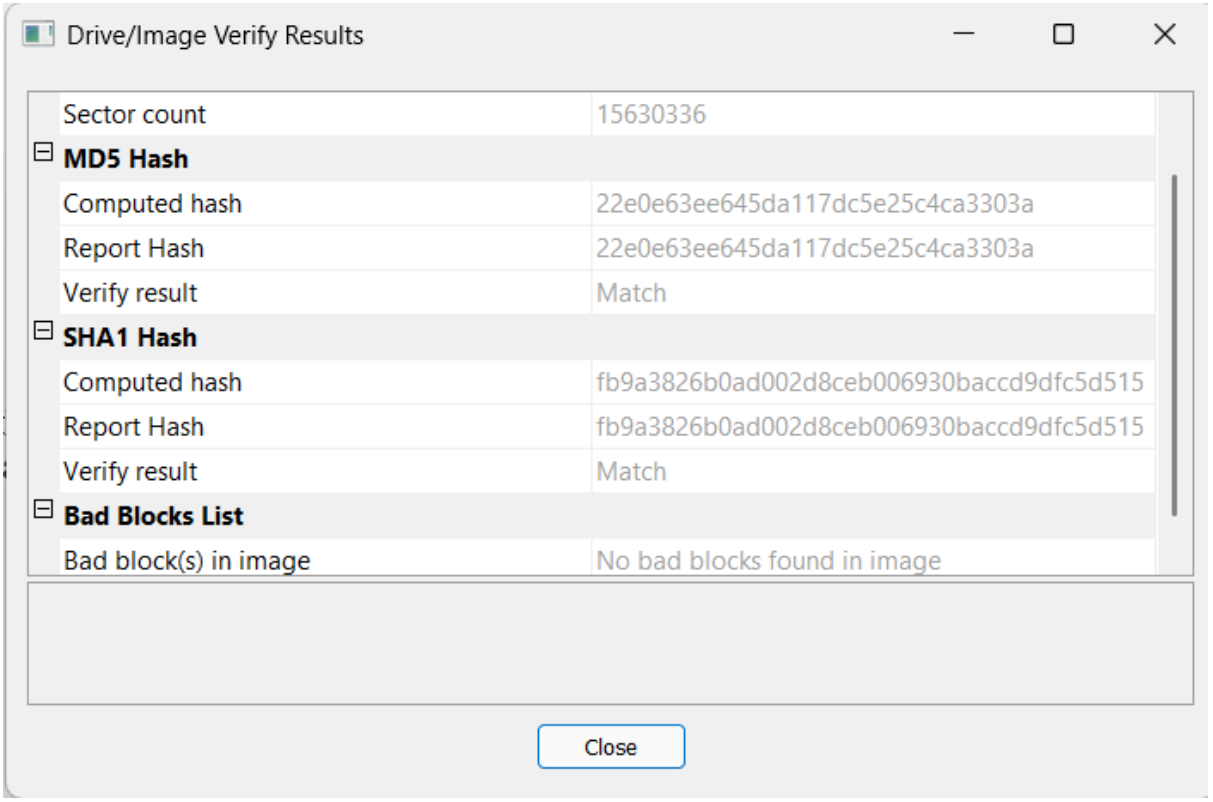
Finally, add the destination of the image file, name the image file, and then click on Finish.



Wait for a few minutes for the image to be created.



After the image is created, a hash result is generated which verifies the MD5 Hash, SHA1 Hash, and the presence of any bad sectors.



Lab Tasks & Questions:
SECTION A: CONCEPTUAL TASKS

Task 1: Logical vs Physical Disk Imaging

What distinguishes a logical disk image from a physical disk image in FTK Imager?

Answer:

Physical disk image:

A physical image is a bit-by-bit copy of the entire storage device. It includes the filesystem, unallocated space, slack space, partition table, and boot records.

Logical disk image:

A logical image captures only selected files, folders, or logical volumes as seen by the operating system. It does not include unallocated space or low-level disk structures.

Why would you choose to acquire a logical disk image instead of a physical one?

Answer:

A logical image is chosen when:

- Only specific files are relevant to the investigation

- Time and storage constraints exist
- Full disk access is not permitted
- Live system acquisition is required without disrupting operations
-

Task 2: Image Integrity Verification

Did you verify the integrity of the image? If so, how did you perform the verification?

Answer:

Yes, the integrity of the forensic image was verified using cryptographic hash functions. SHA-256 hashes were calculated for the source device and the acquired forensic image file

Matching hash values confirmed that the image was an exact bit-by-bit copy and had not been altered during acquisition. Hash verification ensures authenticity, integrity, and forensic soundness of the evidence.

Task 3: DD Command for Disk Imaging

What is the purpose of using the dd command for disk imaging, and how does it compare to other imaging tools like FTK Imager?

Answer:

Comparison:

1. dd:
    a. Command-line based
    b. Produces raw (DD) images
    c. Requires manual hashing and documentation
    d. High risk if used incorrectly
2. FTK Imager:
    a. GUI-based
    b. Supports RAW and E01 formats
    c. Performs automatic hash calculation
    d. Safer and more user-friendly

dd is lightweight and universally available, while FTK Imager is preferred in professional environments for usability and built-in validation.

Task 4: Guymager Disk Image Formats

What types of disk image formats does Guymager support, and how did you choose the most appropriate format for your case?

Answer:

Guymager supports the same types as dd:

- RAW (dd): Simple bit-by-bit copy, widely compatible
- E01 (Expert Witness Format): Compressed, metadata-rich, widely accepted
- AFF: Advanced forensic format with metadata support

The reason for format selection was -

- RAW/DD format is chosen for simplicity, transparency, and compatibility with multiple forensic tools.
- E01 is preferred when compression and metadata documentation are required.

Task 5: Documentation and Screenshots

Provide screenshots of the disk image you obtained, along with supporting documentation that includes details about the tool used and the type of acquisition (physical or logical) performed.

Answer:

Please refer to this document above

Task 6: Insights and Importance

What insights did you gain from completing the disk acquisition tasks in this lab?

Answer:

- Accurate device identification is important
- Unmounting media before acquisition prevents contamination problem
- Hash verification is mandatory for forensic validity
- Tool choice depends on OS, scenario, and investigation scope

Why is the process of acquiring and creating disk images crucial in digital forensics, and how does it ensure the integrity and validity of digital evidence?

Answer:

It is important as it:

- Preserves original evidence in a read-only state
- Enables repeatable analysis without altering evidence
- Maintains chain of custody
- Ensures legal admissibility of digital evidence

SECTION B: TECHNICAL QUESTIONS (As per the forensic image created by the student)

1) 1. 2. 3. 4. 5. What was the size/capacity of the original media as identified by FTK Imager?
   Answer: Source data size: 7632 MB

2) How many sectors were on the original media as identified by FTK Imager?
   Answer: Number of sectors depends on sector size. From FTK we see it to be 15630336

3) What was the serial number of the original drive as identified by FTK Imager?
   Answer:

- On macOS using dd, the serial number was not extracted
- On windows with FTK we got the serial number
  Drive Serial Number: 03025027031924144550

4) What were the MD5 and SHA1 hash values of the original evidence file?
   Answer:
   MD5 - 22e0e63ee645da117dc5e25c4ca3303a
   SHA1 - 1fb9a3826b0ad002d8ceb006930baccd9dfc5d515

5) Was the forensic image an exact copy of the original media?
   Answer: Yes, The forensic image was verified to be an exact copy as the cryptographic hash values of the source and the acquired image matched exactly.